

Executive Summary

An investigation of employee resignation trends and workforce characteristics for a workforce of 100,000 workers is presented in this paper. The purpose of the analysis was to evaluate the workforce structure, look at the differences between employees who left and those who stayed, find out if attrition was concentrated within specific employee segments, and ascertain whether the workforce data that was available could offer trustworthy evidence for comprehending or forecasting resignation.

10,010 workers, or 10.01% of the workforce, quit. Resignation rates remained relatively constant across departments, job titles, gender, education level, frequency of remote work, age, tenure, compensation, and employee satisfaction notwithstanding this degree of attrition. Effect-size analysis and statistical testing also revealed no significant differences between retained and resigned workers in any of the continuous workforce metrics that were looked at.

Multivariate analysis has reinforced these conclusions. The logistic regression model showed very limited explanatory power, but the predicted performance of invisible employees was weak. The remaining test data produced a **ROC-AUC of about 0.492**, indicating that the model cannot reliably distinguish between the resigned and the remaining employees. At the standard 0.50 classification threshold of the standard, the model achieved **89.99%** accuracy, but this resulted from the classification of all employees as retained, resulting in zero accuracy, recall and F1 scores for the resignation group. The conclusions therefore do not support the use of available workforce characteristics as a reliable employee-level resignation prediction tool.

Moreover, they point out that the current dataset does not contain sufficient information to identify the main reasons for resignation. From a management point of view, evidence does not support broad retention interventions targeted solely at certain departments, demographic groups, salary categories, tenure groups or satisfaction categories. On the contrary, the analysis highlights the need for better workforce intelligence through structured data collection, employee experience measurement, continuous detriment monitoring and the integration of additional information such as career progression, manager relationships, organisational culture, benefits and reasons for leaving. The analysis ultimately provides a scientifically based assessment of what current labour data can explain and where there are important information gaps. Resolving these gaps will create a stronger basis for future workforce planning, retention strategies, and responsible predictive analysis.

Business Context and Objectives

Employee retention is an important consideration for workforce management, as employee departures can affect workforce continuity, planning, recruitment requirements and organisational capacity.

This analysis was conducted to examine the characteristics of the workforce and to assess whether the available employee data could provide meaningful evidence of the patterns of resignation.

The analysis had five main objectives:

- **Establish the overall structure and characteristics of the workforce.**

- **Compare employees who have resigned to those who have remained in order to identify significant differences in their workforce characteristics.**
- **Examine deprivation in key employment segments to determine whether resignation was concentrated within certain groups.**
- **Assess the statistical relationship between labor characteristics and resignation.**
- **Evaluate whether the available variables support reliable forecasts for employee dismissals.**

The analysis combines descriptive analysis, statistical tests, evaluation of effects, correlation analysis, logistic regression, classification analysis and validation outside the sample. Throughout the analysis, the observed relationships were treated as associations rather than causal relationships.

Consequently, these findings reflect the evidence contained in the available data sets and should not be interpreted as a determination that a single characteristic of the workforce causes a resignation of an employee.

Workforce Overview

100,000 workers from various departments, job titles, demographic groups, educational backgrounds, and remote work arrangements are included in the dataset. Overall, the workforce shows little variation in organisational makeup and is fairly evenly distributed across departments and job titles. While educational attainment is concentrated among employees with bachelor's and high school degrees, gender representation is also fairly balanced. A wide range of employment characteristics, such as age, tenure, pay, working hours, workload, overtime, training, promotions, performance, and employee satisfaction, are captured in the dataset. Comparatively little variation was found across these workforce measures according to departmental analysis. Departments were largely comparable in terms of average tenure, pay, working hours, projects completed, overtime, training, performance, and satisfaction. For the ensuing attrition analysis, this comparatively stable workforce profile offers helpful context.

Attrition and Workforce Findings

The workforce noted an overall **resignation rate of 10.01%**, with 10,010 out of 100,000 employees identified as having resigned. The analysis revealed minimal evidence that resignations were focused within particular employee groups. Resignation rates were generally similar across various **departments, job positions, genders, education levels, and frequency of remote work**. Chi-square tests revealed no significant statistical relationship between these workforce segments and resignation, and the Cramér's V values suggested a weak association strength.

Additional division by **age, duration of employment, income, and job satisfaction** yielded similarity steady resignation rates. None of these segments showed a statistically significant relationship with resignation.

A direct comparison of resigned and retained employees revealed very comparable average values in **age, tenure, salary, working hours, projects managed, overtime, sick leave, training, promotions, performance, and employee satisfaction**. Testing of independent samples revealed no statistically significant differences between the two groups, with effect sizes being minimal.

The combined descriptive and statistical data suggests that resignation was **not significantly distinguished by the general workforce attributes present in the dataset**. This indicates that additional in-depth employee experience and organisational data might be necessary to comprehend that root causes linked to workforce turnover.

Statistical and Predictive Analysis

Statistical analyses were used to determine whether the characteristics of the observed workforce were significantly related to the resignation of employees.

The correlation analysis found that the relationship between continuous workforce variables and resignations was very weak. None of the correlations between resignation and resignation are statistically significant. The strongest correlation between the workforce's characteristics is between **monthly wages and performance scores**, and the correlation is about **0.51**.

The logistic regression model provided further evidence of limited explanatory value. The model produces **McFadden pseudo R-squared** values of **0.000399** and **probability ratio test p-values of 0.837**, which show very limited improvements compared to models without predictors.

Model discrimination is also weak. The ROC-AUC was approximately **0.516 in the total data set** and **0.492 in the waiting test data**. The average precision is approximately **0.104 for the complete data set**, compared to 10.01% for the resignation base and **0.098 for the tests still in progress**.

At the standard classification threshold of 0.50, the model classified all employees as retained. This produced an accuracy of **89.99%**, but resulted in **zero precision, recall and F1 scores for resigned employees**. This proves that accuracy alone will provide an inaccurate assessment of model effectiveness, considering the imbalance between the resigned and retained employees.

Overall, statistical and prediction analysis shows that available labour variables provide **limited evidence to explain or predict employee resignation**. This model should not be used in its current form as an employee-level resignation risk tool.

Key Business Implications

The analysis carries significant implications for how the organisation ought to handle workforce management and retention.

First, **the 10.01% resignation rate should be regarded as a baseline for monitoring the workforce, not as proof of a particular retention problem or its cause**. The analysis does not pinpoint any specific department, demographic group, salary band,

tenure group, or satisfaction category with evidence strong enough to warrant a targeted intervention.

Second, the fact that resignation rates are consistent across the workforce segments examined indicates that **broad employee characteristics on their own cannot explain workforce attrition**. Management should therefore be careful about basing retention decisions purely on demographic or employment characteristics.

Third, the logistic regression model's weak predictive performance shows that **predictive analytics should not be adopted merely because a model can generate an apparently high accuracy score**. Here, the 89.99% accuracy at the default threshold masked the model's failure to identify employees who had resigned. Out-of-sample validation offered stronger evidence that the model did not generalise well.

Finally, the findings bring to light an important **gap in workforce information**. Variables such as career progression, manager relationships, organisational culture, employee experience, benefits, internal mobility, and structured reasons for leaving are not adequately captured in the available data. Without these factors, the organisation has only a limited ability to investigate the underlying reasons behind resignation.

The key management implication, therefore, is to **strengthen the organisation's workforce intelligence before putting in place highly targeted retention strategies or employee-level predictive systems**.

Recommendations

Strengthen Employee Exit and Experience Data

The limited explanatory and predictive power of the available variables indicates that important factors associated with resignation may not be captured in the current workforce dataset.

HR should strengthen the collection of structured exit information, including reasons for leaving, manager relationships, career-development opportunities, compensation competitiveness, workload concerns, organisational culture, benefits, and external employment factors.

Employee-experience surveys should also be incorporated into workforce analytics where appropriate. This would allow future retention analysis to move beyond basic demographic and employment characteristics and provide a more complete view of the employee experience.

Monitor Attrition Continuously Rather Than Targeting Broad Segments

Since resignation rates were generally similar across the workforce segments analysed, wide-ranging retention strategies aimed solely at specific departments, demographic groups, salary ranges, or tenure categories do not have strong support from the evidence.

HR ought to set up consistent attrition tracking by utilising department, position, tenure, pay, workload, performance, satisfaction, and various pertinent employee-experience metrics. Alterations in these metrics over time can subsequently be examined when significant variations occur.

This method would enable management to react to new workforce threats based on present data instead of depending on conjectures about which employee categories are more prone to leave.

Develop a More Complete Workforce Analytics Framework

The analysis demonstrates the value of combining descriptive, statistical, and predictive methods, but it also highlights the limitations of relying on a narrow set of employee characteristics.

A more complete workforce analytics framework should integrate HR records with **employee engagement measures, career progression, manager and team information, benefits, absenteeism, performance history, internal mobility, and structured exit data**, where available.

Integrating these sources would provide a broader representation of the employee experience and create a stronger evidence base for future analysis of workforce stability, retention, and workforce planning.

Use Predictive Models Only When Sufficient Evidence Exists

The logistic regression model demonstrated limited discriminatory and predictive performance, particularly on the held-out test data. It should therefore **not be used as an employee-level resignation risk scoring mechanism in its current form**.

Future predictive modelling should only be introduced when richer and more informative workforce variables become available and when models demonstrate acceptable performance on unseen data.

Any future employee-risk model should also be subject to appropriate **validation, monitoring, fairness assessment, and governance** before being considered for operational HR decision-making.

Establish Evidence-Based Retention Reviews

HR leadership should use the current **10.01% resignation rate as a baseline for ongoing workforce monitoring**, rather than treating it as evidence of a specific underlying cause.

Periodic retention reviews should examine whether resignation rates, employee-experience measures, and workforce conditions are changing over time. Where material changes occur, targeted investigation can then be undertaken using both quantitative workforce data and qualitative employee feedback.

This approach reduces the risk of implementing broad retention initiatives without sufficient evidence and creates a stronger foundation for future workforce planning.

Management Implication

The principal management implication is that the organisation should strengthen its workforce intelligence before making highly targeted retention decisions.

The analysis does not identify a single demographic, department, compensation level, tenure group, or employee characteristic that can reliably explain resignation. Instead, it demonstrates the importance of improving the breadth and quality of workforce information available to HR.

Management should therefore prioritise **better employee-experience measurement, structured exit-data collection, continuous attrition monitoring, and responsible use of predictive analytics**. This would create a stronger evidence base for identifying emerging workforce risks and designing targeted interventions when meaningful patterns become visible.

Limitations

The results of this analysis need to be viewed in light of the data that was available and the analytical scope of the project.

The main limitation is that the dataset covers a fairly narrow range of employee and employment characteristics. Although it covers demographic, compensation, workload, performance, training, and satisfaction measures, it does not sufficiently capture factors such as career progression, manager relationships, organisational culture, employee engagement, benefits, internal mobility, labour-market opportunities, or structured reasons for leaving. These factors may matter for understanding resignation, but they cannot be assessed directly from the data at hand.

The analysis also relies on a cross-sectional workforce dataset. Consequently, it offers only limited evidence on how employee characteristics and resignation patterns shift over time. The analysis therefore cannot establish temporal relationships or determine whether shifts in workforce conditions come before resignation.

Even though statistical tests and multivariate modelling were applied, the analysis uncovers associations rather than causal relationships. The lack of a statistically significant relationship in this dataset does not prove that a workforce characteristic has no effect on employee retention in other organisational settings.

The predictive modelling results also come with significant limitations. The logistic regression model showed weak discriminatory performance, especially on unseen data. It should therefore not be treated as a dependable operational model of resignation risk. Future predictive analysis would need richer workforce data, suitable validation, and continuous model governance.

Lastly, the dataset reflects a particular workforce and measurement framework. As a result, the findings should not be automatically extended to other organisations, industries, or employee populations without further evidence.

These limitations do not mean the analysis is invalid. Instead, they define the boundaries of what can reasonably be concluded from the information available and point to ways to strengthen future workforce analytics.

Conclusion

The analysis offers a thorough evaluation of workforce structure, resignation patterns, statistical relationships, and the predictive value of the employee data that is available.

The workforce showed a resignation rate of 10.01%, yet resignation was spread broadly across the employee characteristics that were examined. No strong statistical or practical evidence emerged to suggest that resignation was concentrated within specific departments, demographic groups, tenure categories, salary bands, or satisfaction groups.

The statistical and predictive analyses arrived at a consistent conclusion. The workforce variables available displayed only limited association with resignation, and the logistic regression model showed weak explanatory and out-of-sample predictive performance. As a result, the model should not be used in its current form to make resignation-risk decisions at the employee level.

The key finding is that the organisation's capacity to understand resignation is limited by gaps in the workforce information available. Improving exit-data collection, employee-experience measurement, workforce monitoring, and the integration of broader HR information would create a stronger foundation for future analysis.

In the end, the analysis shows the value of relying on evidence rather than assumptions to guide workforce decisions. Instead of naming a single group as responsible for attrition, the findings direct management towards enhancing the quality and breadth of workforce intelligence required to understand employee retention and support more effective workforce planning.